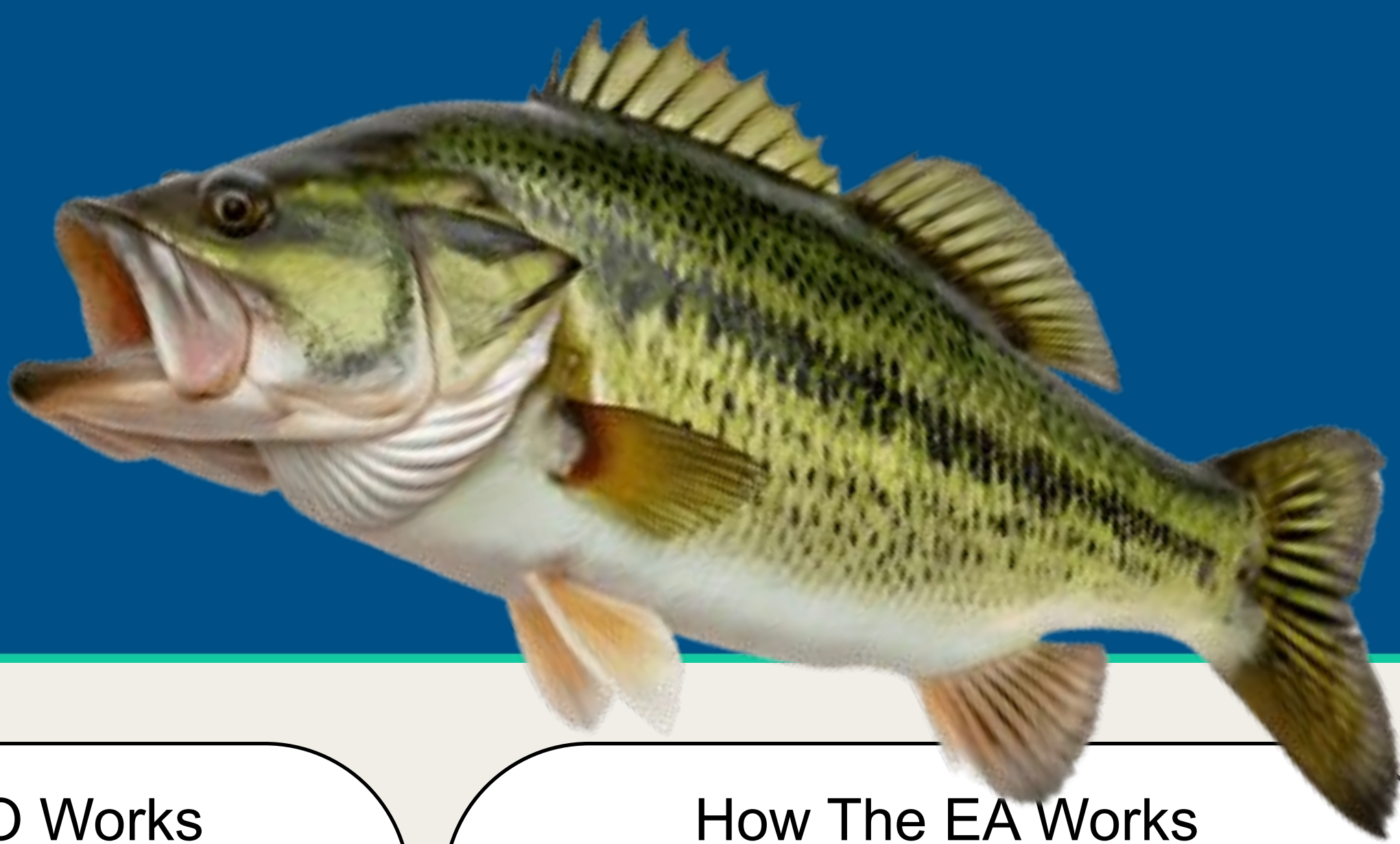


AquaOptima

Applying PSO and EA to simulate and optimize Largemouth Bass Aquaculture

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Core Idea

AquaOptima is an aquaculture simulation that pairs Particle Swarm Optimization to model how a swarm of Largemouth Bass survives a 3D pond, with an Evolutionary Algorithm that selects the best ponds with policy that yields the best outcome.

Background

Largemouth bass aquaculture is a growing global industry [1, 4] facing environmental hurdles like poor winter survival [2] and biological threats such as external columnaris [3].

To combat disease and improve yields, farmers often use dietary supplements to boost fish immunity, metabolism, and growth [6]. A uniquely fascinating challenge in cultivating this species is its highly predatory nature, which frequently leads to cannibalism [5].

Driven by constraints in young (Age 0) fish, this cannibalistic behavior requires active swarm management and optimal environment policies to ensure population health and an increasing population in the pond[5].

How The PSO Works

The particles in the swarm adjust their movement and eating behavior to increase survival depending on these parameters.

cFood 3x Toward nearest food source when fullness < 100%	cProbiotic 1.5x Toward nearest probiotic when immunity < 100%
cOxygen 2x Toward nearest oxygen bubble when < 70%	cSocial 1.5x Toward global optima
cSelfish 1.5x Away from nearest fish	cNH3 4x Away from nearest poison area
cDisease 2.5x Away from nearest disease area	cParasite 3x Away from nearest parasite area
cObstacle 2x Away from nearest obstacle	cRelief 4x Toward obstacle/wall to scrub off parasite
cRun 5x Away from nearest larger mouthed fish	cWander 1x Toward pond center, random when no neighbors

How The EA Works

Pond Genotype

Encodes the aquacultural policy, defining maximum initial fishes, quantity, interval, and drop-location for Food, Probiotic and Oxygen.

Selection Method

Preserves top best using Elitism (k = 20%) unchanged. Remaining parents selected via Tournament (k = 40%) to balance diversity and selective pressure.

Generic Variation

Introduces novelty through 50% independent Crossover and a 25% Mutation rate, randomizing genes to explore the globally optimal solution.

Fitness Evaluation

Quantifies performance through weighted sum of biological yield, economic savings, and average swarm health.

Fitness Score Formula

$$\text{Fitness} = 0.55 \times \text{Yield} + 0.33 \times \text{Saving_Rate} + 0.12 \times \text{Healthiness}$$

• Yield (55%): Aim to produce maximum living fish
• Saving_Rate (33%): Spend as little as possible
• Healthiness (12%): Survivors should be in good condition

Primary Flow Diagram

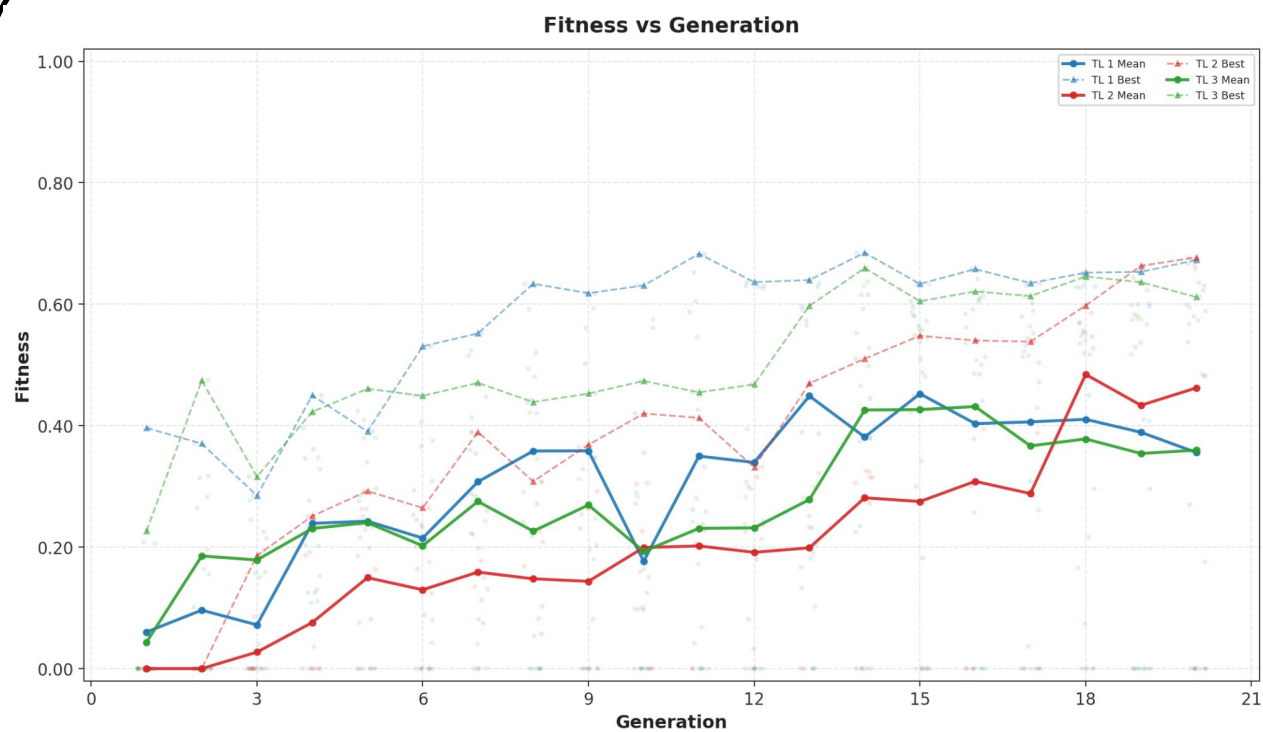
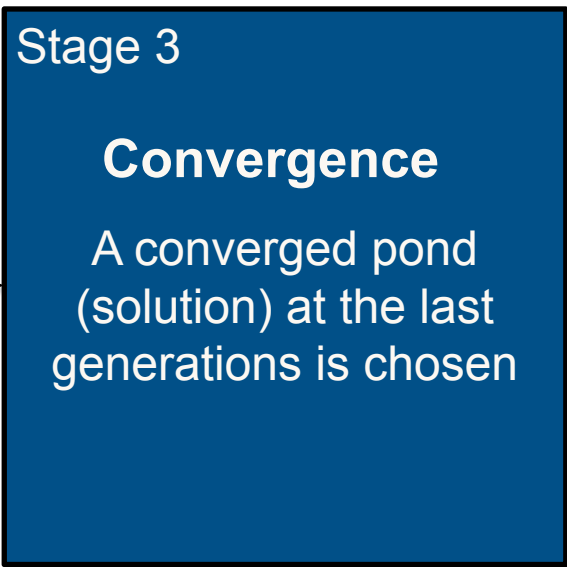
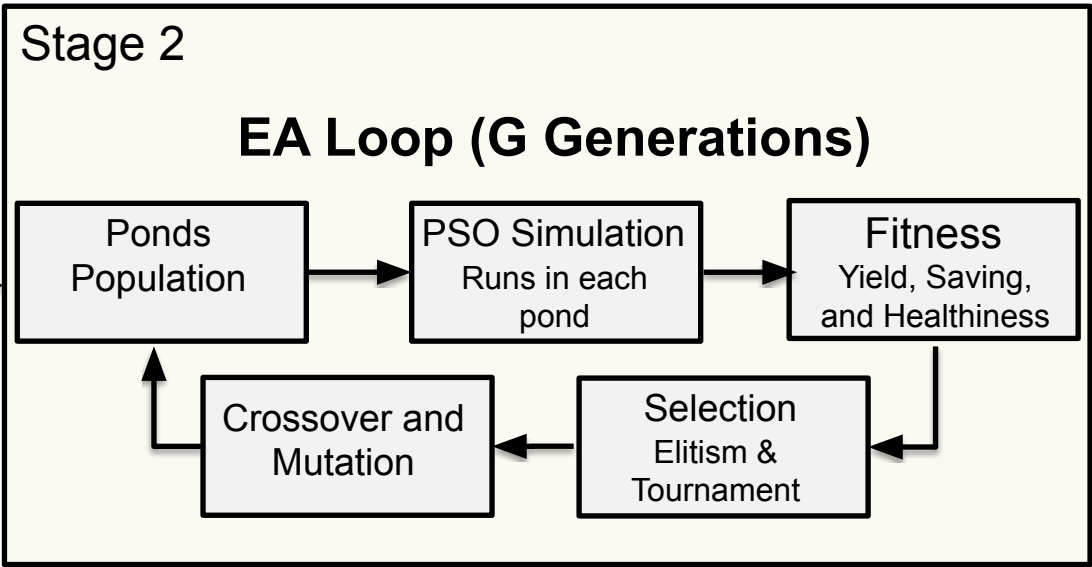
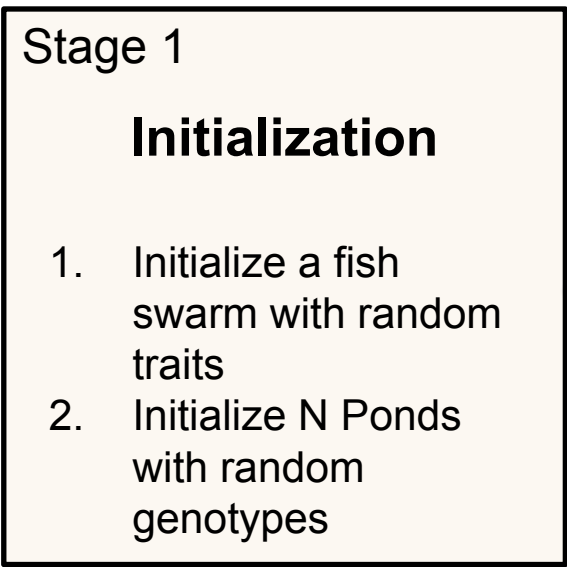


Fig. 1: Fitness vs Generation: Shows average and peak fitness scores improving across the generations

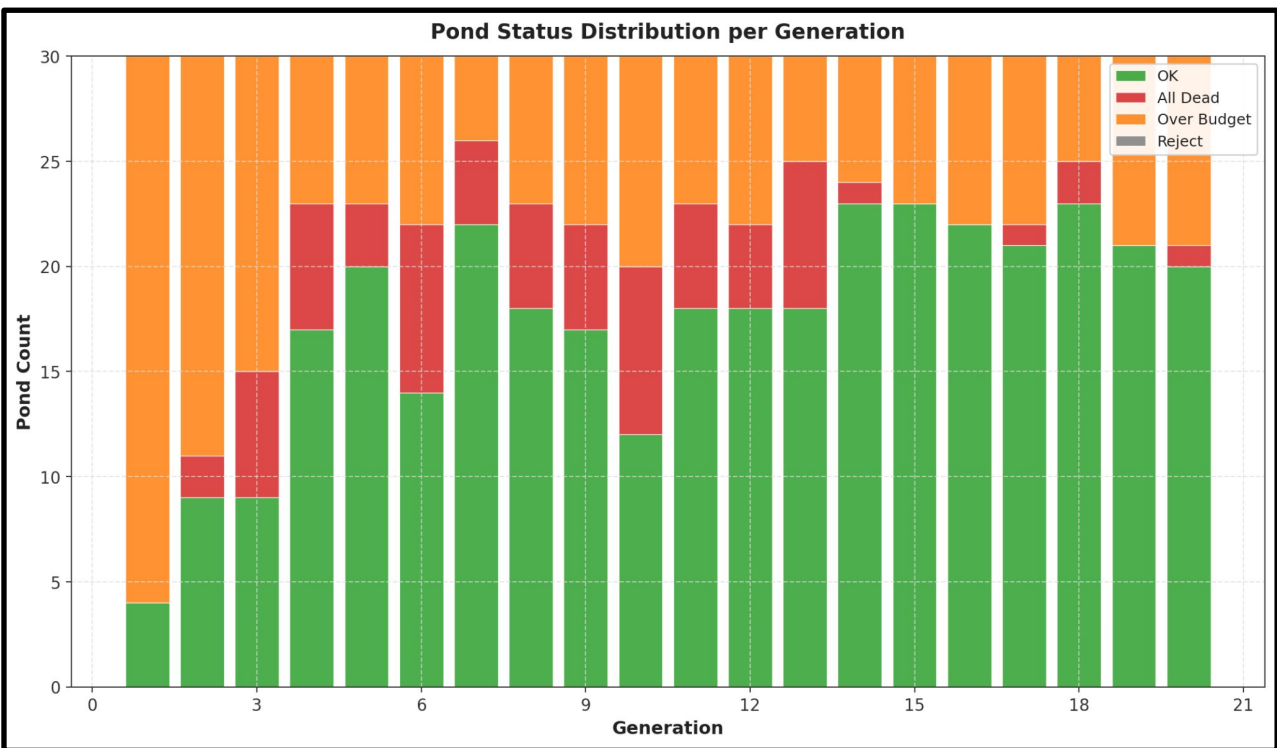


Fig. 2: Pond Status Distribution: Count the final states of the ponds

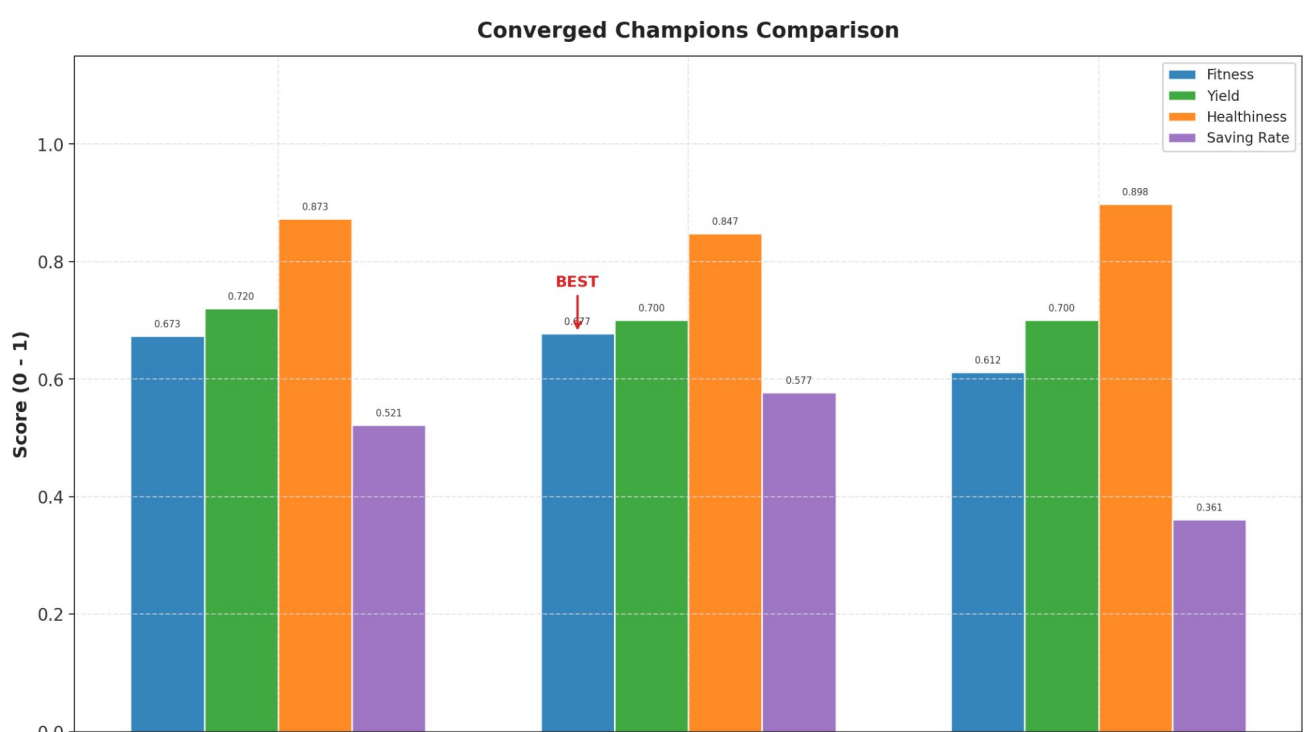
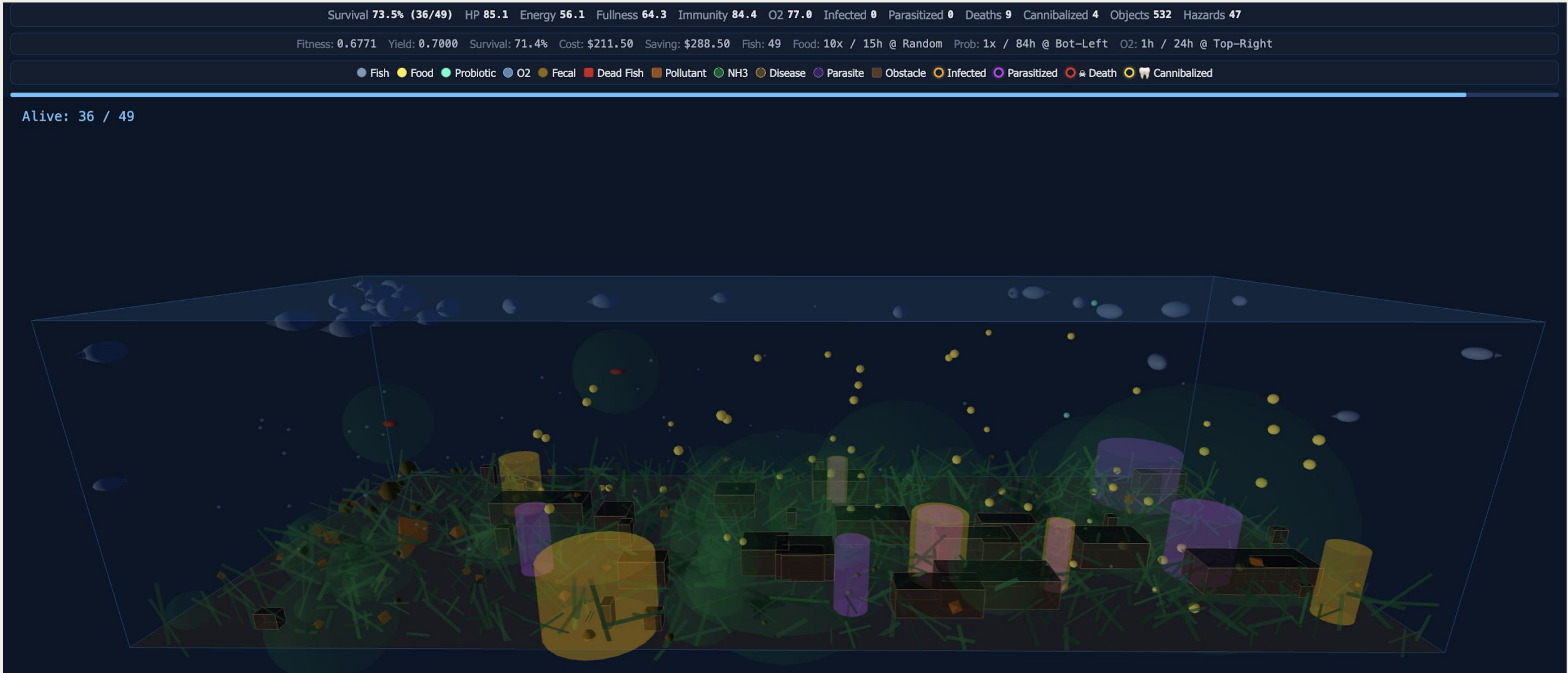


Fig. 3: Converged Champions: Compares the final fitness metrics



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